Akshit Kartik

(814) 441-5652 • axk6143@psu.edu • github.com/AKAAKS • linkedin.com/in/akshit-kartik

Education

The Pennsylvania State University, University Park, PA **Expected May 2026**
Bachelor of Science in Computer Science GPA: 3.56
Minor: Cybersecurity Computational Foundations
Relevant Coursework: Data Structures & Algorithms, Operating Systems, Computer Security, Databases, Web Development, Object-Oriented Programming

Work Experience

Meritus.AI ***Chennai, India***
Research and Development Intern **June 2024 - August 2024**

- Developed and deployed YOLO-based object detection models onto the CV Pro AI Kit, utilizing OpenCV and Python to identify and track competition elements for the World Robot Olympiad (WRO) 2024.
- Engineered firmware in C++ for Arduino and ESP-32 microcontrollers to process sensor inputs, manage motor control, and execute real-time pathfinding logic for autonomous navigation.

The Pennsylvania State University ***University Park, PA***
Undergraduate Researcher **January 2025 - May 2026**

- Build the data pipeline for a CNN-LSTM gesture correction model, using OpenCV to preprocess video clips and Pandas to label and timestamp kinematic errors in sign language for downstream model training.
- Develop a custom Python annotation GUI for the AquaBit swimmer-tracking platform, synchronizing IMU sensor and video streams with NumPy/Pandas to produce a labeled multimodal dataset for activity recognition.

The Pennsylvania State University ***University Park, PA***
Learning Assistant **January 2024 - December 2025**

- Lead weekly recitation sections for 40+ students in a large-scale Java course, teaching object-oriented programming, data structures, and web application fundamentals.
- Provide targeted support by grading 100+ programming assignments, leading office hours, and delivering constructive feedback to improve code quality and design patterns.

Project Experience

AI-Powered Parts Procurement Engine **October 2025 - December 2025**

- Built a full-stack Next.js 15 / React 19 / TypeScript web application for **B. Braun Medical Inc.** with NextAuth.js closed-list authentication, integrating the OpenAI API to automate sourcing of obsolete manufacturing components and replace a manual workflow that previously took weeks to months per part.
- Designed a multi-field search and part-comparison engine that normalizes vendor specifications (manufacturer, model, voltage, dimensions) and exports vetted matches to Excel via SheetJS (xlsx); the final prototype successfully matched 100% of items in B. Braun's obsolete-part test catalog and reduced per-part search time to 1–2 minutes.

OS Agent — Autonomous Windows System Optimizer **January 2026 - Present**

- Architected a Python agent that continuously monitors CPU, RAM, GPU, and processes via psutil and triages anomalies through a LangGraph-orchestrated OODA (Observe–Orient–Decide–Act) state machine backed by a 4-layer decision engine and SQLite behavioral history.
- Integrated OpenAI GPT models with confidence-based escalation and a ChromaDB vector store of past fixes, eliminating redundant LLM calls on recurring anomalies and enforcing a configurable per-session cost cap.
- Hardened the agent with a human-in-the-loop approval gate, command allowlisting and sanitization, protected-PID guards, and a Docker sandbox for risky actions; verified with 73 pytest unit tests across the decision, safety, and tooling layers.

HealthCare Insurance Claim Fraud Detector **August 2024 - Present**

- Developed fraud detection models (Logistic Regression, Random Forest) using Python with Scikit-learn, Pandas, and NumPy to analyze and classify large-scale insurance datasets.
- Engineered features, resolved severe class imbalance using SMOTE, and optimized model hyperparameters with GridSearchCV, achieving a validated F1-score of 0.78 on the minority fraud class.

Technical Skills

- **Languages:** Python, Java, C, C++, JavaScript, TypeScript, SQL, HTML/CSS, Bash
- **Frameworks & Libraries:** Next.js, React, Node.js, NextAuth.js, LangGraph, PyTorch, TensorFlow/Keras, Scikit-learn, YOLOv8, OpenCV, Pandas, NumPy, Tailwind CSS
- **Tools & Platforms:** Git, GitHub, Docker, OpenAI API, ChromaDB, SQLite, REST APIs, Linux/Ubuntu, Windows, Arduino, ESP-32